

are licensed to use the Bluetooth technology and the Bluetooth brand and logo at no charge. This allows an open standard which can reach out to a broad market base. Inter-compatibility with various other electronic devices also allows for mass consumer acceptance of this technology.

How Bluetooth works: Bluetooth uses radio frequency (RF) signals to establish a point-to-point and point-to-multipoint voice and data transfer. The term that best explains the networks created is a personal area network. Bluetooth-enabled devices within a range of 30 feet (10 metres) form a piconet, an ad hoc wireless network where communication occurs via RF signals. For a device to communicate with another, both must contain a Bluetooth radio. The radio, which is extremely small, is actually built into a computer chip which manages the individual connections.

The radio also consumes very little power. Every Bluetooth radio conforms to the exact same specifications for both transmitting and receiving

Bluetooth was developed by Ericsson to connect earphones to your mobile phone without wires

signals, so that they can be used anywhere in the world without modification. Bluetooth devices can also link through a series of piconets to larger local area networks, and the global internet.

Bluetooth simplified connections across electronic devices for two reasons. One - wireless connectivity. Two - it initiated a global standard with every Bluetooth device following similar specifications and communication protocols. Be it with wires or without, connecting two different devices made by different manufacturers can be quite a complex procedure. The simplest example is that of the mobile charger, different companies have different pins (though there is talk of a soon to be released universal charger). In the wired world, creating the proper physical connection is sometimes quite challenging. Plugs, pin to pin connectivity, various connectors - serial, parallel, optical and so on - the choices are mind boggling. Beyond that, there's serial communication where data is sent one bit at a time, and parallel processing, where data is processed in groups of 8 or 16 bits. The Bluetooth specification therefore bypasses all these troubles, and this is one of the main reasons for its widespread